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Experiment 7: Logistic Regression (LR)

Aim

To implement the **Logistic Regression (LR)** algorithm in Python and classify data samples.

Python Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=200)

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

print("Predicted Values:")
print(y_pred)
```

```
print("\nActual Values:")
print(y_test)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Predict new sample
new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)

print("\nNew Sample Prediction:")
print(iris.target_names[prediction[0]])
```

Sample Output

```
...  Predicted Values:
     [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
      0 0 0 2 1 1 0 0]

     Actual Values:
     [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
      0 0 0 2 1 1 0 0]

     Accuracy:
     1.0

     Confusion Matrix:
     [[19  0  0]
      [ 0 13  0]
      [ 0  0 13]]

     New Sample Prediction:
     setosa
```